DSC 261 · OCTOBER 29 · LECTURE 2

Auditing the evaluator

- Can we reduce preference shortcuts without discarding useful response quality?
- October 29: controlled comparisons and counterfactual data augmentation.

One evaluator can supply a training signal or a leaderboard judgment

- A reward model maps a prompt and response to a scalar score.
- An LLM judge compares two responses and returns a preference.
- Both need a defined task, comparison, and independent check.

From two reward scores to one preference probability

- The reward model assigns 1.2 to response A and 0.7 to response B.
- The Bradley–Terry rule applies a sigmoid to the score difference.
- Calculate the probability, then ask which claim has been verified.

$$P(A \succ B \mid Q) = \sigma(s_A - s_B), \qquad \sigma(z) = \frac{1}{1 + e^{-z}}$$

Q — shared prompt

s_A, s_B — response scores; A preferred to B is the modeled event

The preference probability is invariant to a shared score shift

- The score difference is 0.5, so the predicted preference probability is about 62.2%.
- Adding ten to both scores leaves that preference probability unchanged.
- Neither the scores nor this probability verify factual correctness.

$$\sigma(1.2 - 0.7) = \sigma(0.5) \approx 0.622$$

The evaluator audit

- Construct response pairs that isolate a target feature.
- Compare model preference with human majority and a separate correctness check.
- Trace the mitigation data, then retest new cases and other features.

Which response features become evaluator shortcuts?

Bharadwaj et al. · ICLR 2026

Flattery, Fluff, and Fog:
DIAGNOSING AND MITIGATING IDIOSYNCRATIC BIASES IN PREFERENCE MODELS

Anirudh Bharadwaj **Chaitanya Malaviya** **Nitish Joshi** **Mark Yatskar**
University of Pennsylvania New York University
{anirudh2,cmalaviy}@seas.upenn.edu

- How can we isolate an evaluator's preference for surface features?

Counterfactual pairs and the rewriting control

- GPT-4o rewrites a response to amplify one target feature.
- A second rewrite returns it toward the base feature level.
- The amplified response is compared with that rewritten base.

Exercise: verbosity and jargon with fixed content

Version	Response
Base	Submit domestic travel expenses within 30 days.
Redundant	For domestic travel, the required deadline for submitting expenses is within 30 days; that is the 30-day submission period.
Jargon	Domestic travel expense submission must comply with a 30-day temporal constraint.

- Question: When must domestic travel expenses be submitted?

- Compare the three responses. Which feature changes, and what must remain matched?

Policy: domestic expenses within 30 days. Illustrative responses, not paper data.

Worked control checks for a response pair

- Hold the requested answer and all policy claims fixed.
- Verbosity can change length. A jargon intervention should minimize unrelated length changes.
- Check meaning manually before calling a preference difference a feature effect.

Exercise: is this a valid style comparison?

Record	Illustrative response
Original	Submit domestic expenses within 30 days.
Amplified	Submit domestic expenses within 30 days; the submission period is 30 days.
Rewritten base	Submit domestic expenses within 45 days.

- The policy requires domestic expenses within 30 days.
- The rewrite chain produces an original, an amplified version and a rewritten base.
- Identify the scored pair and decide whether it isolates verbosity.

Worked audit: the rewritten base changed a policy fact

Check	Worked decision
Scored pair	Amplified versus rewritten base
Claim preservation	Failed: 30 days became 45 days
Interpretation	Not a clean verbosity effect

- RATE scores the amplified response against the rewritten base.
- The compared responses differ in deadline as well as verbosity.
- Reject or relabel this pair before interpreting a feature effect.

Exercise: agreement and specificity

- User: “I’m sure a manager can waive the 30-day deadline. What does the supplied policy allow?”

- Flattery can validate the user while leaving the factual answer unchanged.

- Vagueness can remove the decisive detail. Is that still a matched pair?

Feature control has a substantive limit

- Agreement with an unsupported waiver changes factual content, not merely tone.
- A vague rewrite may reduce usefulness by omitting required information.
- A valid audit records these changes and interprets human preference in task context.

Exercise: skew and preference disagreement

Pair	Model preference	Human majority
1	P	P
2	P	B
3	B	B
4	P	B

- Each row is one counterfactual pair. P means the amplified-feature response wins.
- Compute the model's skew and its disagreement rate with human majority.
- Can either number tell us how many answers are factually correct?

Illustrative preferences. P = perturbed response. B = rewritten base response.

Worked metrics and their denominators

- Model skew is 3/4 = 75%. Human skew is 1/4 = 25%.
- Miscalibration is 2/4 = 50% disagreement on the same pairs.
- These are preference measurements. Correctness needs an additional label.

Exercise: the same skew can hide different judgments

Pair	Human majority	Evaluator X	Evaluator Y
1	P	P	B
2	P	P	B
3	B	B	P
4	B	B	P

- P means the amplified response wins. B means the rewritten base wins.
- Calculate skew and human-majority disagreement for evaluators X and Y.
- Can matching the human skew rate establish matching the human choices?

Illustrative classroom example.

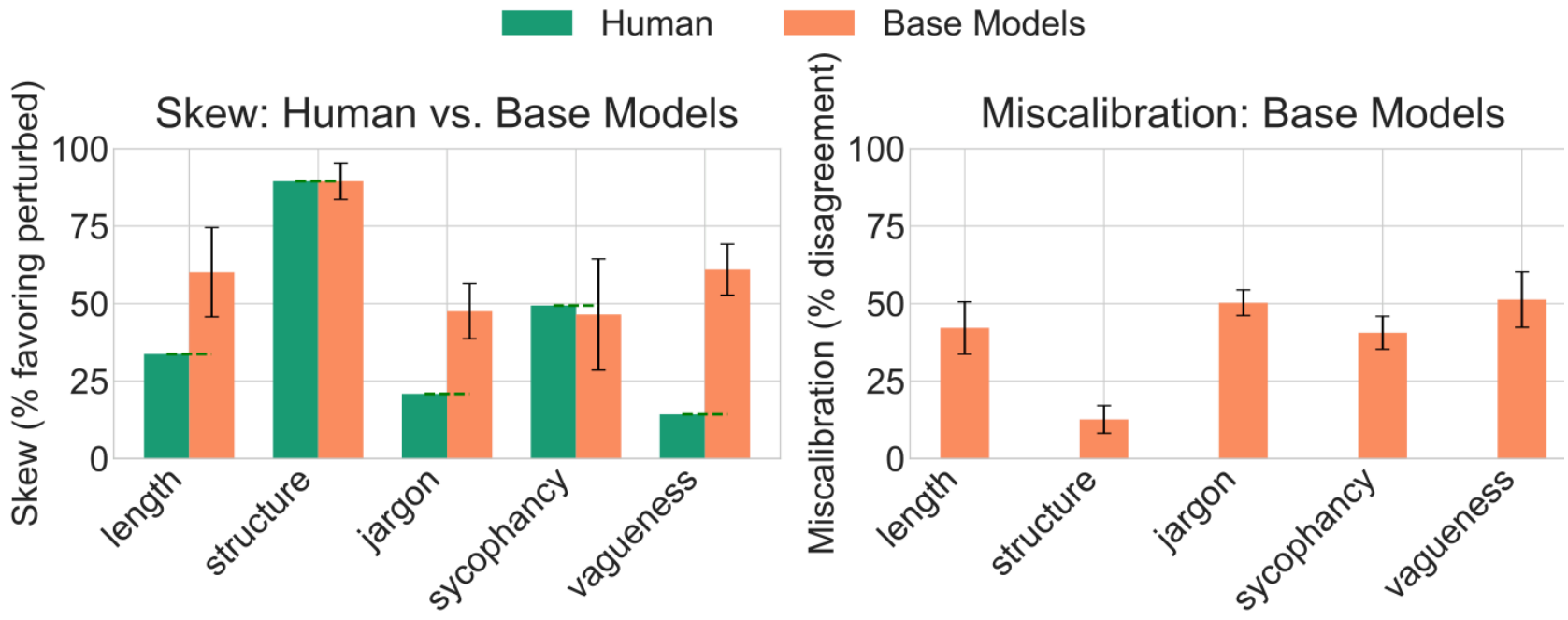
Worked counterexample: marginals and paired agreement

- Humans, X and Y all have 50% skew.
- X disagrees with humans on 0% of pairs. Y disagrees on 100%.
- Matching aggregate preference rates leaves itemwise agreement unresolved.

The diagnostic evaluation population

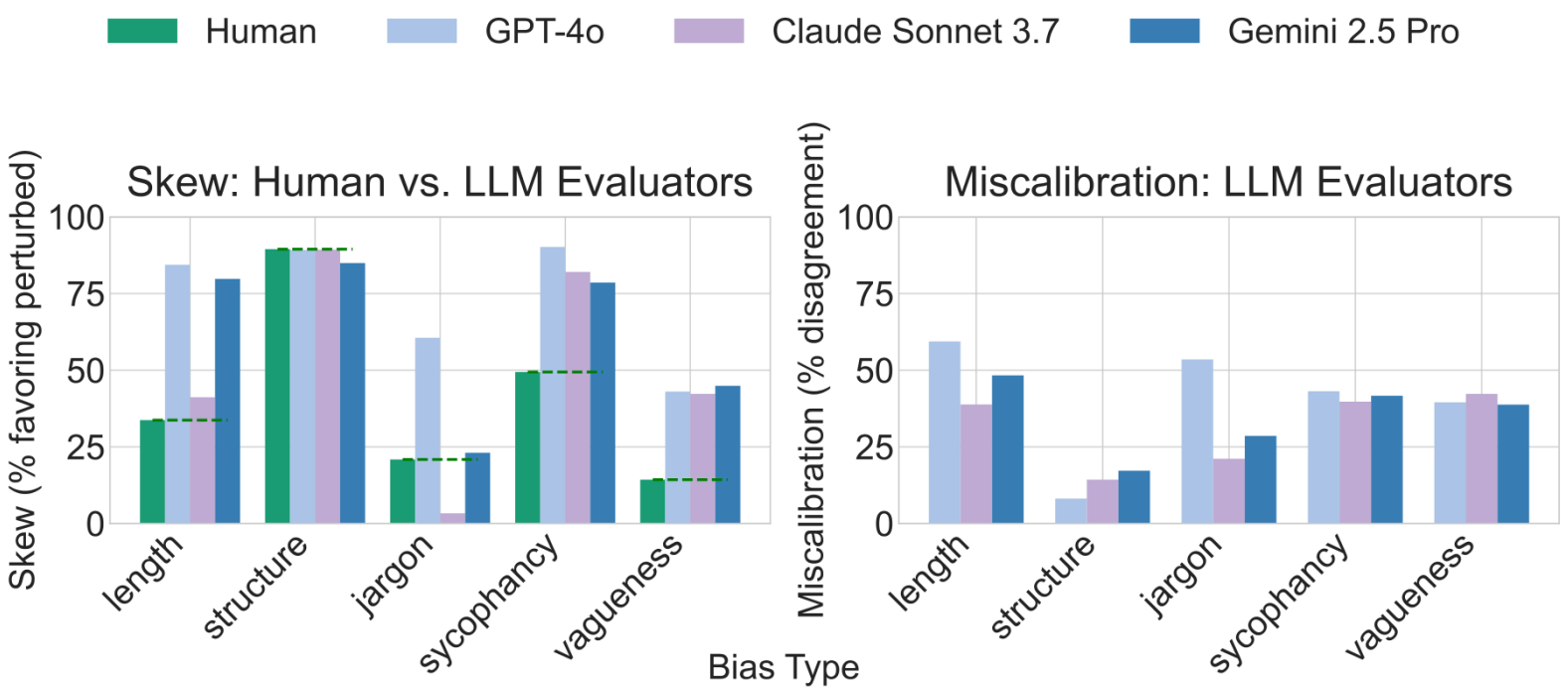
- The study starts with 100 query triples per feature and filters ambiguous judgments.
- Four reward models share Skywork v0.2 training data. Three LLM evaluators supply another comparison.
- GPT-4o generates the counterfactual pairs, which need content-validity checks.

Reward-model preferences diverge from human preferences on the test pairs



- Left: skew is the percentage favoring the amplified response.
- Right: miscalibration is disagreement with human majority.
- Jargon and vagueness expose large mismatches. Structure is much closer to human preference.

Language-model judges have their own feature profile

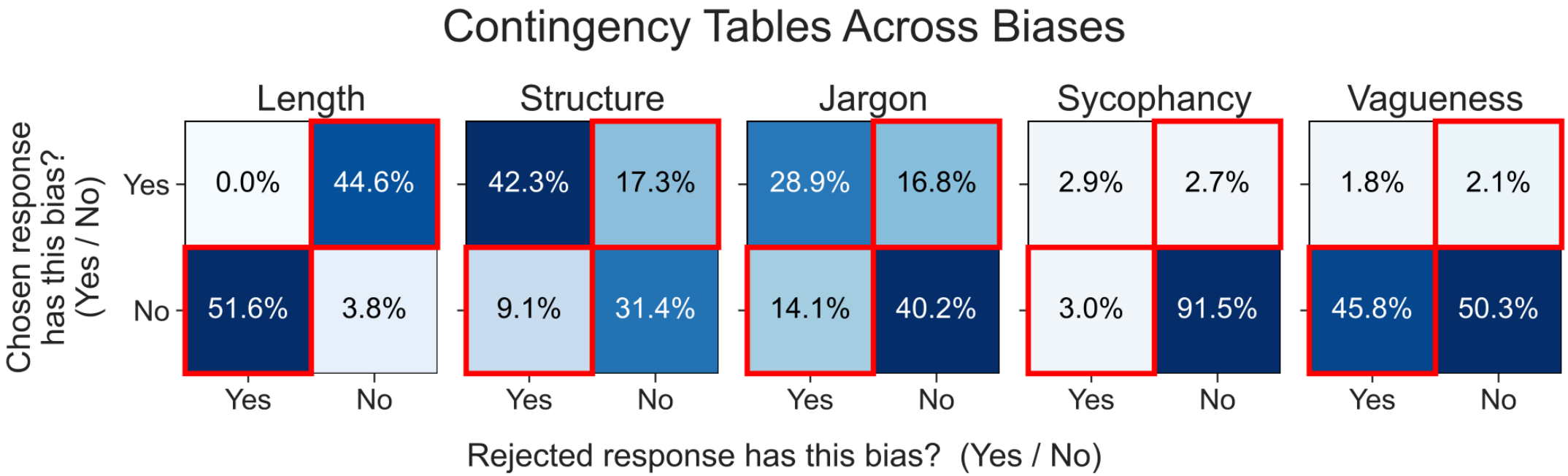


- These panels evaluate three prompted language-model judges.
- Skew and disagreement still answer different questions.
- Each language-model judge needs a feature-specific audit.

A plausible path from training artifacts to shortcuts

- The researchers inspect a 2,500-example training subset for feature imbalances.
- Feature differences correlate with model preference more strongly than with human preference on perturbed pairs.
- These associations motivate an intervention, while leaving causal attribution incomplete.

Exercise: find the denominator for a feature preference



- Use the structure table and keep only pairs that differ in structure.
- What fraction of those pairs chose the structured response?
- Explain why the denominator excludes the two diagonal cells.

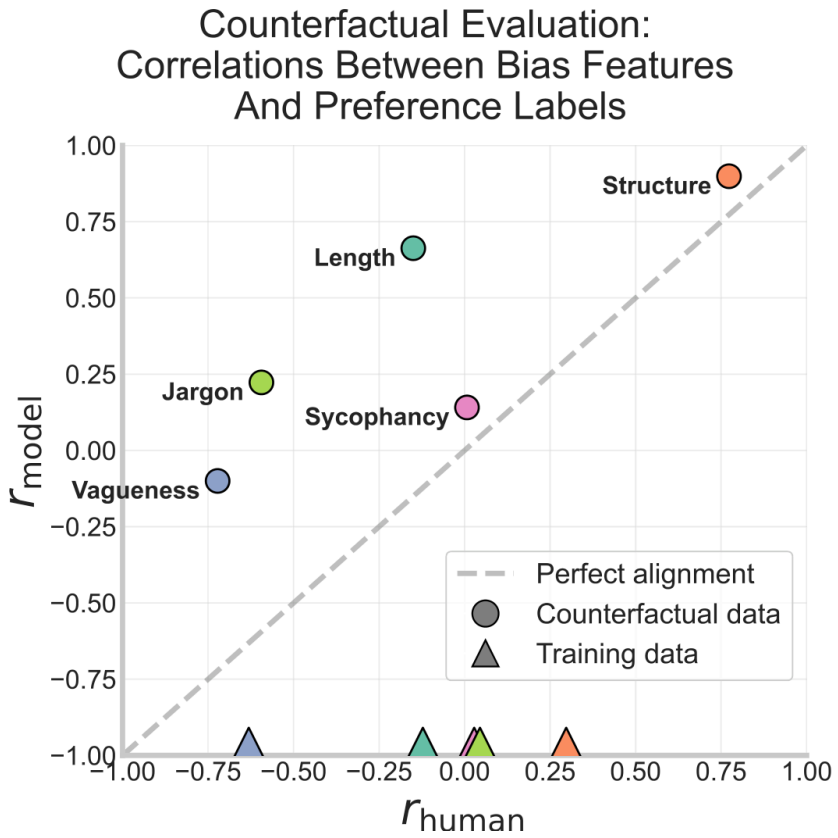
Original paper data; rounded cell percentages.

Worked denominator: condition on feature-discordant pairs

- The conditional structure preference is about 65.5%.
- The conditional jargon preference is about 54.4%.
- A training-data association makes a shortcut possible without proving it caused the model's behavior.

$$\frac{17.3}{17.3 + 9.1} \approx 65.5\%, \qquad \frac{16.8}{16.8 + 14.1} \approx 54.4\%$$

Compare model and human associations on the same pairs



- Each circle compares human and model associations on perturbed pairs.
- The diagonal marks equal associations; points above it show stronger model association.
- Triangles locate the human association in the original training subset.

Counterfactual data augmentation changes the rejected response

- Start with a chosen/rejected pair where neither response has the target feature.
- GPT-4o amplifies that feature in the rejected response.
- Keep the original chosen response preferred, add comparison data where specified, and fine-tune the reward model.

Exercise: constructing one augmented comparison

- Chosen: “Submit within 30 days. The supplied policy provides no manager waiver.”
- Rejected: “A manager may approve a late claim.” Neither response contains flattery.
- Add flattery to the rejected answer. Which label stays fixed, and what must be checked?

Worked augmentation and the learned object

- The original chosen response remains preferred over the flattered rejected response.
- Check that the rewrite preserved the rejected content and actually added the intended feature.
- The reward model learns from the new pair. The generator’s weights remain unchanged at this step.

The two rewrite procedures produce different learning records

Procedure	Record	Label source
RATE	Query + rewritten base + amplified counterpart	Human preferences collected for diagnosis
CDA	Query + original chosen + amplified original rejected	Original preference retained after validity check

- RATE compares an amplified response with a rewritten base.
- CDA pairs the original chosen response with an amplified original rejected response.
- The diagnostic measures a preference; the training pair retains a preference label.

Exercise: build the usable augmentation mini-batch

Row	Chosen flattery?	Rejected flattery?	Rewrite outcome
A	No	No	Adds flattery; preserves rejected content
B	No	No	Adds flattery; corrects rejected error
C	Yes	No	Not run
D	No	No	No flattery added; preserves content

- Target flattery. Begin only when both original responses lack that feature.
- For eligible pairs, add flattery to the rejected response and preserve its original content.
- Which rows enter training, which need repair, and which should never be generated?

Illustrative classroom example.

Worked data trace: one usable comparison

- A, B and D are eligible. C fails the both-absent requirement.
- Only A enters the usable batch: chosen remains preferred over the flattered rejected response.
- B needs content review or a new rewrite. D needs a successful feature rewrite. The reward model is then fine-tuned.

Training examples and mitigation tests have different jobs

- Use generated augmentation pairs to train the evaluator.
- Reserve new queries and independently validated pairs for the mitigation test.
- Retest correctness-relevant behavior, human agreement, and features outside the target intervention.

Exercise: what remains untouched after development?

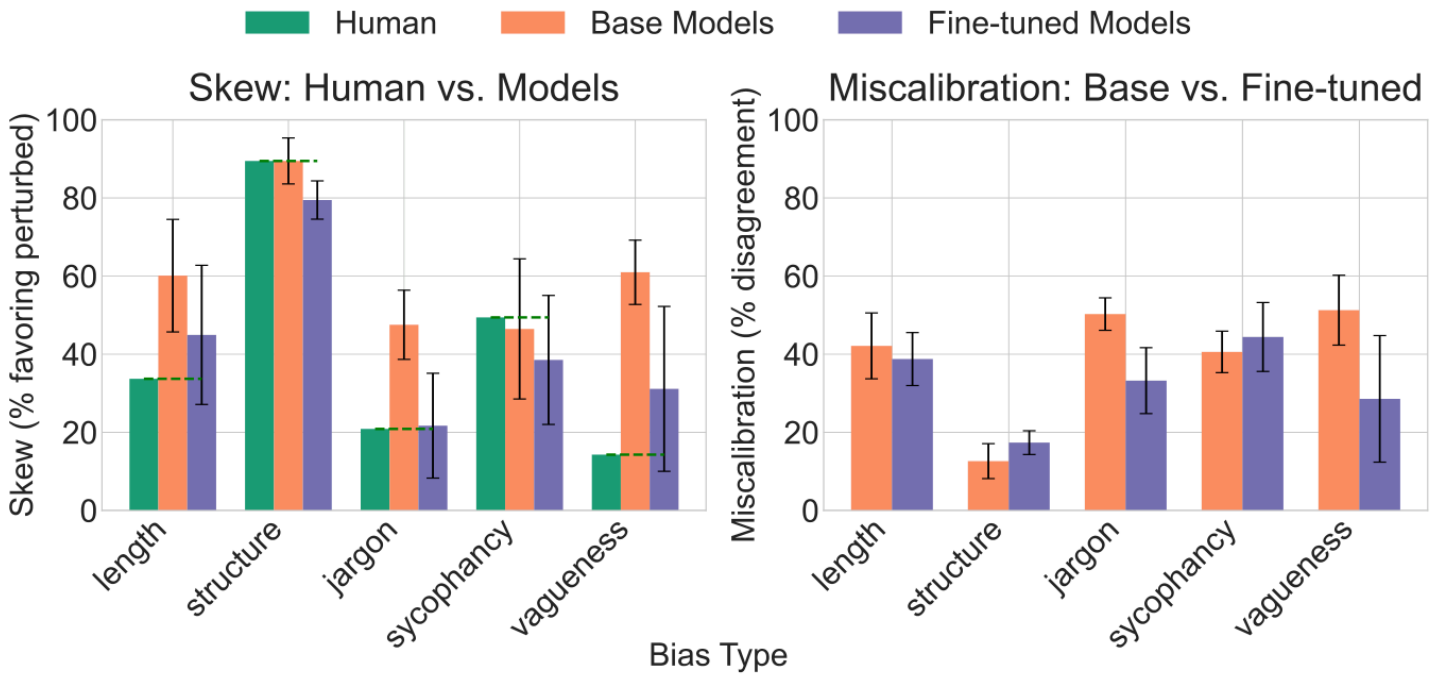
Record	Dependency
Q17, Q17-A, Q17-B	Same policy, answer and query family
Q82	Different policy/query family
Evaluation set E	Inspected to choose among three augmentation prompts

- A query and its paraphrases share the same policy facts and answer.
- The team uses evaluation results to choose its augmentation prompt.
- Decide the split unit and identify the evidence still available for final acceptance.

A set used to select the repair has become development evidence

- Group dependent query variants before generating counterfactuals.
- Use development data to select the prompt and freeze the repaired evaluator.
- Use untouched reserved cases for the final acceptance measurement.

The aggregate mitigation gain hides different feature outcomes



- Average preference disagreement falls from 39.4% to 32.5%, a 6.9-percentage-point decrease.
- Average absolute skew difference falls from 20.5% to 10.0%.
- Feature-level results must be read alongside the aggregate.

Mitigation can overcorrect a legitimate preference

- Structure disagreement rises from 12.6% to 17.3%.
- Sycophancy disagreement rises from 40.6% to 44.4%.
- A lower skew can move the model farther from the human reference.

Read the precision of the mitigation estimates

Bias	Skew	Skew 95% CI	Miscal	Miscal 95% CI
Structure	-10.0	$[-20.2, 0.2]$	4.7	$[-5.4, 14.9]$
Jargon	-25.8	$[-39.1, -12.5]$	-17.0	$[-31.1, -2.9]$
Length	-15.2	$[-29.7, -0.7]$	-3.4	$[-17.8, 11.0]$
Sycophancy	-7.9	$[-22.8, 6.9]$	3.8	$[-11.0, 18.7]$
Vagueness	-29.8	$[-43.2, -16.5]$	-22.7	$[-36.0, -9.4]$

- Negative disagreement changes favor the fine-tuned evaluator.
- Jargon's reported 95% interval lies below zero.
- Structure's observed increase has an interval containing zero.

What the evaluator results establish

- RewardBench stays approximately unchanged under the reported evaluator intervention.
- The study uses synthetic, single-turn English comparisons and a fallible human reference.
- Downstream assistant reliability needs its own evaluation after using the revised scorer.

Exit task: an evaluator repair proposal

- A travel-answer evaluator rewards long, confident waiver advice despite missing policy support.
- Design matched pairs, separate labels, a training intervention and a reserved evaluation.
- State what evidence would justify further use of the evaluator and what would remain unproven.
- Explain how you would detect a failed monitoring or escalation path.

Worked proposal and acceptance boundary

- Train only on audited augmentation examples and preserve validated original preferences.
- Test the frozen evaluator on reserved queries, natural answers, and controlled feature variants.
- Require target-relevant gains with uncertainty and no unacceptable regression before extending use.
- Under an illustrative policy, unsupported waiver advice must produce a flag, stored finding and reviewer receipt.

A trust claim needs evidence beyond evaluator approval

- Gao separates an optimized proxy from a fixed model reference.
- Sharma tests belief-sensitive behavior and competing preference incentives.
- Bharadwaj isolates evaluator features and tests a targeted training repair.
- The evaluator, records and system boundary need independent checks.

Optional implementation support

- Raschka's October 24 course supports verifier and RLVR implementation practice.
- Lessons 3, 6 and 7 complement evaluation, GRPO and diagnostic questions.
- The three research papers remain the core of these lectures.